PAVEL CHERKASHIN

Software Engineer • Agent Harness & Distributed Systems • Managed Agents Platform

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Operating Focus: *"Architecting the runtime platform behind real-time multimodal voice agents. Bridging low-latency transport engineering, autonomous agent harnesses (fleetd daemon, slot concurrency, zero-trust git handoffs), and multi-tier evaluation suites (cross-model audit loops, programmatic layout assertions, latency regression tracking) to deliver sub-second, resilient voice agent deployments at production scale."*

CORE TECHNICAL CAPABILITIES & AGENT HARNESS SUBSTRATE

- **Agent Harnesses & Daemon Runtimes:** Custom multi-agent orchestrator architecture (fleetd Rust daemon), deterministic slot scheduling (WIP=1), proactive backlog vacuuming, zero-trust git handoffs, and process isolation.
  - **Conversational Latency Optimization:** End-to-end tracing across STT → LLM → TTS pipelines; profiling Time-To-First-Token (TTFT), streaming chunk delivery, socket buffers, and turn-taking latency.
  - **Evaluation Suites & Multi-Agent QA:** Multi-tier cross-model evaluation gates (Gemini, Codex, Claude), programmatic layout/containment assertions (PyMuPDF/pypdf), automated regression testing, and verification protocols.
  - **Deployment Resilience & Systems Architecture:** Architectural focus on high-throughput streaming backends, zero-downtime rolling upgrades, fault-tolerant message queuing, and distributed service reliability.
  - **Frontier Coding-Agent Velocity:** Fluent daily operation across modern agentic loops (Claude Code, Gemini CLI, OpenAI tools); disciplined spec-driven development, sharp PR reviews, and automated CI/CD gates.

SYSTEMS ENGINEERING & AGENT HARNESS EXPERIENCE

Mindrock Autonomous Agent Mesh — *Principal Systems Architect & Harness Builder* | 2024 – Present

- **fleetd Daemon & Slot Concurrency:** Engineered the monolithic fleet daemon in Rust, coordinating distributed agent slots across macOS nodes (wolfram, darwin) with autonomous task-vacuum loops, rate-limiting, and strict WIP caps.
  - **Protocol Phi & Zero-Trust Coordination:** Created sealed JSON envelope transport (proto: phi-0.1) across Git repositories, enforcing cryptographically verifiable handoffs, isolated workspace lifecycles, and tamper-proof SHA-256 receipts.
  - **Continuous Evaluation & QA Gate Architecture:** Built automated multi-tier QA test suites (cross-agent verification between Gemini, Codex, and Claude) executing programmatic layout assertions, functional unit tests, and validation scripts prior to Principal human gate approval.
  - **Real-Time Execution Telemetry:** Deployed structured telemetry logging capturing task claim timestamps, execution durations, agent exit codes, and automated error recoveries across distributed fleet nodes.

Vestor.in & High-Growth Venture Platforms — *Founding Partner & Technical Systems Architect* | 2018 – 2024

- **Distributed Systems Architecture:** Architectural focus on event-driven platforms, API gateway resilience, and high-concurrency data streaming across portfolio technology platforms.
  - **Resilient Transport & Integrations:** Technical leadership on distributed webhook/websocket integrations, retry mechanisms with exponential backoff, and robust service availability.
  - **Performance & Observability Focus:** Technical guidance on latency profiling, distributed queue contention monitoring, and database scalability across high-growth systems.

Enterprise Systems & Technology Leadership — *Siebel Systems, Sputnik Labs, Microsoft* | Prior

- Country Manager at Siebel Systems; Co-Founder & CEO of Sputnik Labs (pioneer CRM systems integrator in Russia/CIS, scaling to 110+ engineers and specialists, acquired by Technoserv).
  - Platform Director at Microsoft; author of university textbook *Customer Relationship Management Strategy* (■■■■■■ / ■■■■■, 2007, 380 ■., ISBN 978-5-94774-643-3); systems design mentor.

ROLE ALIGNMENT, TRANSPARENT GAPS & OPERATING POSTURE

- **Intentional IC Backend Focus:** Experienced founder/operator deliberately pursuing an individual contributor backend systems role to build foundational, high-leverage agent runtime infrastructure for Cartesia's Managed Agents.
  - **First-Principles Ramp on Voice Protocols:** Direct mastery of agent harnesses, distributed queuing, and streaming sockets; transparently ramping on low-level WebRTC/SIP/RTP telephony protocols and voice frameworks (Pipecat/LiveKit).
  - **Engineering Discipline & Collaborative Presence:** Deeply committed to in-person San Francisco collaboration, sharp technical specifications, rigorous peer design reviews, and high-velocity shipping culture.

EDUCATION & SYSTEMS FOUNDATION

- **B.S. Computer Science in Global Ecology:** Lomonosov Moscow State University (MSU), 1995 (Computational analysis of satellite data, mathematical modeling; research fellowship at IIASA Austria).